MH1101 Chapter 1: Integrals

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1.1 Antiderivatives

Reference: Stewart Calculus 8E, Section 3.9

> **Goal:** There are many antiderivatives of a given function, but they are all the same up to adding a constant.
> We introduce the notation $\int f(x)\,dx$ of an indefinite integral. It is a convenient notation when speaking about antidifferentiations.

When we differentiate a function, we get its (unique) derivative. When we "reverse" the process of differentiation, we get an antiderivative.

Definition 1. (Antiderivative & Indefinite Integral)

A function $F(x)$ is an **antiderivative** of $f(x)$ on an interval (a, b) if

$$F'(x) = f(x) \quad \text{for all } x \in (a, b).$$

$f(x) \rightarrow \frac{d}{dx} \rightarrow f'(x) = \frac{df}{dx}$ (unique)

anti-derivative

$F(x) \leftarrow \frac{d}{dx} \leftarrow f(x)$

not unique

Example 1.1. The following are obvious from the definition (assuming all functions are defined on an appropriate interval):

anti derivatives   derivatives

- f is an antiderivative of f'. $f \rightarrow \frac{d}{dx} \rightarrow f'$
- f' is an antiderivative of f''. $f' \rightarrow \frac{d}{dx} \rightarrow f''$

Example 1.2. Let $f(x) = 3x^2$, where $x \in \mathbb{R} = (-\infty, \infty)$. Consider the functions

$$F(x) = x^3, \quad G(x) = x^3 + 1, \quad H(x) = x^3 + 100.$$

not unique!

By definition, all $F(x)$, $G(x)$, $H(x)$ all antiderivatives of $3x^2$ on $\mathbb{R}$ since

$$F'(x) = G'(x) = H'(x) = 3x^2 = f(x).$$

$\square$

Indeed, in the preceding example, any function of the form $x^3 + C$, where C is a constant, is an antiderivative of $3x^2$. The question is: Are there any others? The answer is No!

The proof below uses the **Mean Value Theorem (MVT)**, which we now recall from Calculus I: Suppose f is continuous on $[a, b]$ and is differentiable on (a, b). Then there exists a point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Theorem 1. All antiderivatives of f differ by a constant. That is, if F is an antiderivative of f on an interval (a, b), then the most general antiderivative of f on (a, b) is

$$F(x) + C$$

where C is an arbitrary constant.

Proof. Let $F(x)$ and $G(x)$ be two antiderivatives of f; so by definition, we have

$$F'(x) = f(x), \quad G'(x) = f(x).$$

We want to show that $G(x) = F(x) + C$ for some constant C. showing that their difference is a constant

Approach: Let $h(x) = G(x) - F(x)$. We will show that $h(x) = C$ for some constant C. To do this: Let x_1, x_2 be two arbitrary numbers inside the interval (a, b), where $x_1 < x_2$. We will prove that $h(x_1) = h(x_2)$ using the Apply the Mean Value Theorem.

$\exists c$ st $h'(c) = \frac{h(x_2) - h(x_1)}{x_2 - x_1}$

Note that

$$h'(x) = G'(x) - F'(x) = f(x) - f(x) = 0 \text{ for all } x \in (a, b). \tag{1.1}$$

Applying the Mean Value Theorem to h on the interval $[x_1, x_2]$, we get a number $c \in (x_1, x_2)$ such that

$$h(x_2) - h(x_1) = h'(c)(x_2 - x_1).$$

Since $h'(c) = 0$ (by (1.1)), we have $h(x_2) - h(x_1) = 0$, whence $h(x_2) = h(x_1)$.

arbitrary

Therefore, h has the same value at any two numbers x_1, x_2 in (a, b). This means that h is constant on (a, b), i.e. $h(x) = C$ for some constant C.

We have proved that $G(x) - H(x) = C$, as desired. $\square$

Notation. The general antiderivative of f is also called an **indefinite integral** of f, and is denoted by

$$\int f(x)\,dx = F(x) + C,$$

where F is an antiderivative of f, and C is an arbitrary constant.

Table of some useful indefinite integrals

$\int k\,dx = kx + C \quad (k \text{ is a constant})$	$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \ (n \neq -1)$
$\int \sin x\,dx = -\cos x + C$	$\int \cos x\,dx = \sin x + C$
$\int \sec^2 x\,dx = \tan x + C$	$\int \csc^2 x\,dx = -\cot x + C$
$\int \sec x \tan x\,dx = \sec x + C$	$\int \csc x \cot x\,dx = -\csc x + C$
$\int e^x\,dx = e^x + C$	$\int a^x\,dx = \frac{a^x}{\ln a} + C \ (a \neq 1)$
$\int \frac{1}{x}\,dx = \ln\lvert x\rvert + C$	

$\frac{d}{dx}\tan x = \sec^2 x$

Some common properties of indefinite integrals (antiderivatives):

- $\displaystyle\int (f(x) + g(x))\,dx = \int f(x)\,dx + \int g(x)\,dx$
- $\displaystyle\int a \cdot f(x)\,dx = a \cdot \int f(x)\,dx$, where a is a constant.

Example 1.3. Find the most general antiderivative of the function

$$f(x) = \frac{1}{2x} + 5\sin x, \quad x \in \mathbb{R} \setminus \{0\}.$$

<u>*Solution.*</u> Using the table of indefinite integrals, we have

$$\begin{aligned}
\int \frac{1}{2x} + 5\sin x\, dx &= \int \frac{1}{2x}\, dx + \int 5\sin x\, dx \\
&= \frac{1}{2}\int \frac{1}{x}\, dx + 5\int \sin x\, dx \\
&= \frac{1}{2}\ln|x| - 5\cos x + C.
\end{aligned}$$

We cannot apply Theorem 1 directly to the domain $\mathbb{R}\setminus\{0\}$ since it is not an interval of the form (a, b). The function is defined on $\mathbb{R}\setminus\{0\} = (-\infty, 0)\cup(0,\infty)$. We can apply Theorem 1 to each domain $(-\infty, 0)$ and $(0, \infty)$ separately:

$$\int \frac{1}{2x} + 5\sin x\, dx = \begin{cases} \frac{1}{2}\ln|x| - 5\cos x + C_1 & x \in (-\infty, 0) \\ \frac{1}{2}\ln|x| - 5\cos x + C_2 & x \in (0, \infty) \end{cases}$$

can drop for this
if want cos x is +ve.

where C_1 and C_2 are arbitrary constants. □

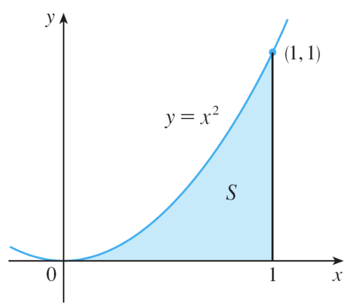
may not be
the same!

1.2 Area Problem

Reference: Stewart Calculus 8E, Section 4.1

Goal: To calculate the area under a curve as the limit of Riemann sums. This motivates the definition of a definite integral in the next section.

Consider the area under the parabola $y = f(x) = x^2$ from 0 to 1:



We approximate the area as follows:

- Divide the interval $[0, 1]$ into n equal subintervals of equal size $\triangle x$:

$$[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n].$$

- Use rectangles with these subintervals as base to approximate the area between the graph and the x-axis.
- Take the values of the function at the **right endpoints** or **left endpoints** of the subintervals as the height of the rectangles.

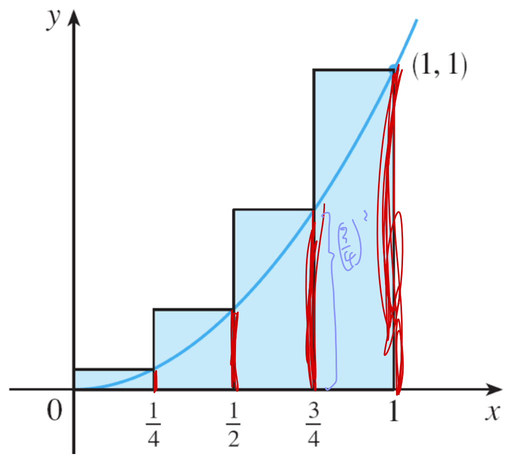
  (i) With right endpoints, the sum of the areas of the rectangles, denoted by R_n, is

$$R_n = \sum_{i=1}^{n} f(x_i) \triangle x.$$

  (ii) With left endpoints, the sum of the areas of the rectangles, denoted by L_n is

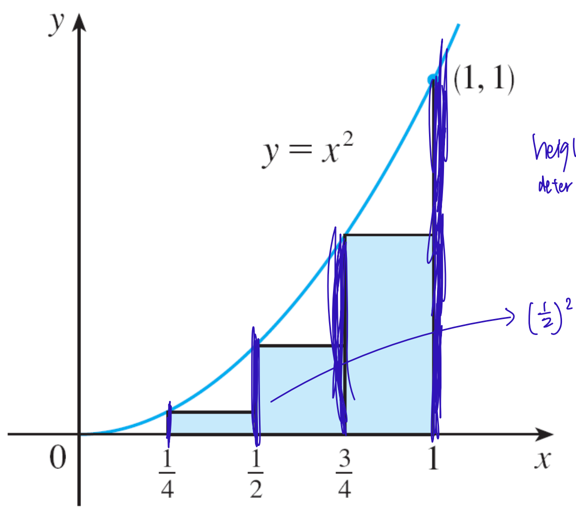
$$L_n = \sum_{i=1}^{n} f(x_{i-1}) \triangle x.$$

Approximation by $n = 4$ subintervals using right endpoints:



height of rectangle
determined by right endpoint
→ overestimation

Approximation by $n = 4$ subintervals using left endpoints:



height of rectangle
determined by left endpoint

Let A be the actual area between the graph of the function and the x-axis.

Let R_n be the approximation of the area using n subintervals with right end-points.

Let L_n be the approximation of the area using n subintervals with left end-points.

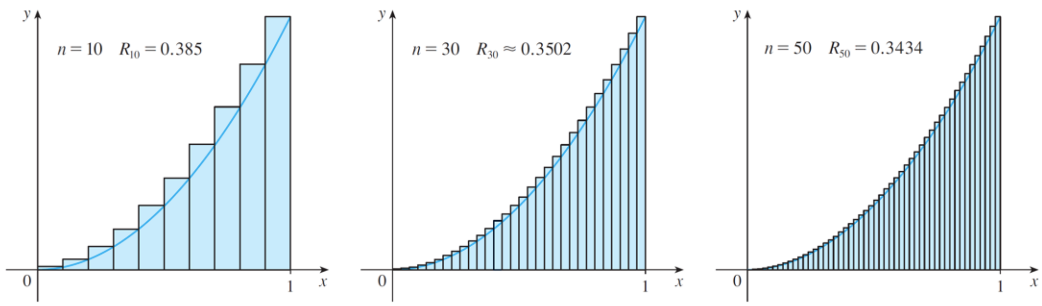
We have

actual area

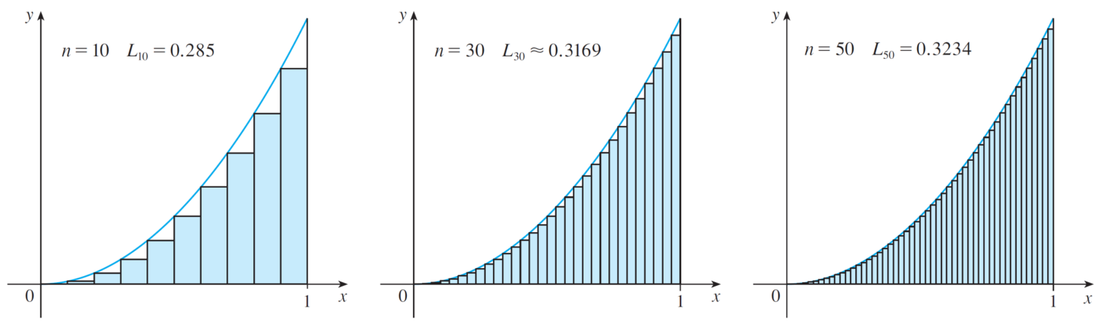
$$L_n < A < R_n.$$

→ not always true
depends if function is increasing or decreasing.

Approximation using right endpoints:



Approximation using left endpoints:



Intuitively, the approximations get better and better as n tends to ∞:

$$A = \lim_{n\to\infty} R_n = \lim_{n\to\infty} \left(f(x_1)\triangle x + f(x_2)\triangle x + \cdots + f(x_n)\triangle x\right).$$

$$A = \lim_{n\to\infty} L_n = \lim_{n\to\infty} \left(f(x_0)\triangle x + f(x_1)\triangle x + \cdots + f(x_{n-1})\triangle x\right).$$

In fact, instead of left endpoints or right endpoints, we could take the height of the i-th rectangle to be the value of f at *any* number x_i^* in the i-th subinterval $[x_{i-1}, x_i]$. We call the numbers x_1^*, x_2^*, ..., x_n^* the **sample points**. So a more general expression for the area is

*can be left or right point, doesn't matter.

$$\begin{aligned} A &= \lim_{n\to\infty} \left(f(x_1^*)\triangle x + f(x_2^*)\triangle x + \cdots + f(x_n^*)\triangle x\right) \\ &= \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\triangle x. \end{aligned}$$

Concepts covered:
① Anti derivatives differ by a constant
② Area = limit of Rn or Ln approximations (Riemann sums)

1.3 Definite Integral

Reference: Stewart Calculus 8E, Section 4.2

Goal: Motivated by the area problem in the preceding section, we define the notion of a definite integral of a function f which occurs in a wide variety of situations even when f is not necessarily a positive function.

Definition 2. (Definite Integral)

Suppose f is a function defined for $a \leq x \leq b$, and we divide the interval $[a, b]$ into n subintervals of equal width $\triangle x = (b-a)/n$:

$$[x_0, x_1], [x_1, x_2], \ldots, [x_{n-1}, x_n].$$

Let $x_i^* \in [x_{i-1}, x_i]$ be any sample points in these subintervals. The **definite integral** of f **from** a **to** b is

limit of Riemann sum

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\triangle x,$$

provided that this limit exists and gives the same value for all possible choices of the sample points.

If the definite integral $\int_a^b f(x)\,dx$ exists, we say that f is **integrable** on $[a, b]$.

The sum $S_n = \sum_{i=1}^{n} f(x_i^*)\triangle x$ is called a **Riemann sum** of f. So the definite integral$\int_a^b f(x)\,dx$ is the limit of the Riemmann sums S_n as $n \to \infty$.

In the notation $\displaystyle\int_a^b f(x)\,dx$:

※ This has nothing to do with differentiation

- $f(x)$ is called the **integrand**.
- a and b are called the **limits of integration**; a is the **lower limit** and b is the **upper limit**.
- The term dx simply indicates that the independent variable is x. It is also a 'dummy' variable in the notation since replacing x by any other variable will not change the value. E.g:

$$\int_a^b f(x)\,dx = \int_a^b f(t)\,dt = \int_a^b f(u)\,du.$$ ⇒ all will give same value

↓

a constant

- The procedure of calculating an integral is called **integration**.

With right endpoints as sample points, we have

$$\boxed{\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\triangle x,}$$

whereas, with left endpoints as sample points, we have

$$\boxed{\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_{i-1})\triangle x,}$$

where

$$x_i = a + i\triangle x \;\; (i = 0, 1, 2, \ldots, n) \quad x_0 = a, \quad x_n = b.$$

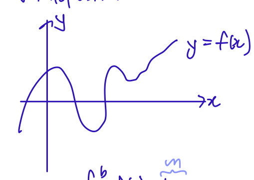
Bad notation:

$\int_a^{x} f(x)\,dx \to$ wrong

$\int_a^{x} f(t)\,dt \to$ correct $\Rightarrow$ ans will be a func. of x (if a is constant and x is variable

$\int_a^{b} f(x)\,dx \to$ correct $\Rightarrow$ ans will be a constant if a and b are constants

Intepretation:



$\int_a^b \underbrace{f(x)}_{\text{mass density kg/m}} \overbrace{dx}^{m}$

$\boxed{\textbf{Example 1.4.}}$ With right endpoints as sample points, compute $\displaystyle\int_0^1 x^2\,dx$ by evaluating the corresponding limit of Riemann sums.

Solution. It is known (by mathematical induction) that

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}.$$

$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

$\sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2$

if needed, will be provided, don't need to memorise

$\int_0^1 x^2\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x$

$= \lim_{n\to\infty} \sum_{i=1}^{n} x_i^2\,\Delta x$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left(\frac{i}{n}\right)^2\left(\frac{1}{n}\right)$

$= \lim_{n\to\infty} \frac{1}{n^3} \sum_{i=1}^{n} i^2 \to$ there's a formula for it.

$\Delta x = \frac{b-a}{n} = \frac{1-0}{n}$

$x_i = a + i\Delta x = 0 + \frac{i}{n}$

Let $f(x) = x^2$. With $a = 0$, $b = 1$, we have

$$\triangle x = \frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}, \quad x_i = 0 + i\triangle x = \frac{i}{n}, \quad i = 0, \ldots, n.$$

With right endpoints as sample points, we have

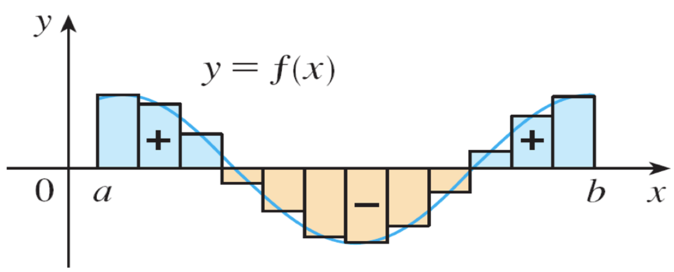
$$\begin{aligned}
S_n &= \sum_{i=1}^{n} f(x_i)\triangle x \\
&= \sum_{i=1}^{n} f\left(\frac{i}{n}\right)\frac{1}{n} = \sum_{i=1}^{n}\left(\frac{i}{n}\right)^2 \frac{1}{n} \\
&= \frac{1}{n^3}\sum_{i=1}^{n} i^2 \\
&= \frac{1}{n^3}\cdot\frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2}.
\end{aligned}$$

Hence

$$\begin{aligned}\int_0^1 x^2\,dx &= \lim_{n\to\infty} S_n \\ &= \lim_{n\to\infty} \frac{(n+1)(2n+1)}{6n^2} \\ &= \frac{1}{6}\lim_{n\to\infty} \frac{n+1}{n}\cdot\frac{2n+1}{n} \\ &= \frac{1}{6}\lim_{n\to\infty}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right) \\ &= \frac{1}{6}\cdot 1\cdot 2 \\ &= \frac{1}{3}.\end{aligned}$$

□

If f takes on both positive and negative values, then the Riemann sum is the sum of the areas of the rectangles that lie above the x-axis and the negatives of the areas of the rectangles that lie below the x-axis (the areas of the blue rectangles minus the areas of the gold rectangles).



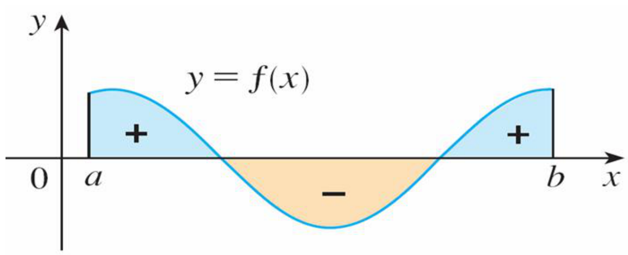
When we take the limit of such Riemann sums, the definite integral can be interpreted as a **net area**, that is, a difference of areas:

$$\int_a^b f(x)\,dx = A_1 - A_2$$

where

- A_1 is the area of the region above the x-axis and below the graph of f (between $x = a$ and $x = b$) ,

- A_2 is the area of the region below the x-axis and above the graph of f (between $x = a$ and $x = b$).



Properties of Definite Integrals

When we defined the definite integral $\int_a^b f(x)\,dx$, we implicitly assumes that $a < b$. In fact, the definition as the limit of Riemann sums still works even if $a > b$, in which case, we would have

$$\triangle x = \frac{b-a}{n} = -\frac{a-b}{n}.$$

Therefore, if we were to switch the limits of integration, we only have to multiply the resulting integral by a '−1'.

$$\boxed{\int_a^b f(x)\,dx = -\boxed{\int_b^a f(x)\,dx}.}$$

Also, if $a = b$, then $\triangle x = 0$ and so

Riemann Sum will be zero!

$$\boxed{\int_a^a f(x)\,dx = 0.}$$

Assuming f and g are continuous and c is a constant, the following basic properties of definite integrals make our life easier:

Basic rules of definite integrals:

(P1) $\displaystyle\int_a^b c\,dx = c(b-a).$

(P2) $\displaystyle\int_a^b (f(x) \pm g(x))\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx.$

(P3) $\displaystyle\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx.$

(P4) $\displaystyle\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx$ $\Rightarrow$ transitive rule

* c doesn't have to be greater than a!

Comparison rules of definite integrals:

(P5) If $f(x) \geq 0$ on $[a, b]$ then $\displaystyle\int_a^b f(x)\,dx \geq 0.$

(P6) If $f(x) \geq g(x)$ on $[a, b]$ then $\displaystyle\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx.$

(P7) If $m \leq f(x) \leq M$ on $[a, b]$ then

$\int_a^b m\,dx \leq \int_a^b f(x)\,dx \leq \int_a^b M\,dx$

follows from P1 and P6

$$m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a).$$

The proofs of the above rules are quite straight forward using the definition of definite integrals. Here we only provide the proofs of Property (P6) and (P7).

Proof of Property (P6). Since $f(x) \geq g(x)$ on $[a, b]$, we have

$$f(x_i^*) \geq g(x_i^*) \text{ for all i} \implies f(x_i^*)\triangle x \geq g(x_i^*)\triangle x \text{ for all i}$$

$\Delta x = \frac{b-a}{n} \Rightarrow$ non neg for $b>a$

$$\implies \sum_{i=1}^n f(x_i^*)\triangle x \geq \sum_{i=1}^n g(x_i^*)\triangle x$$

Taking limit as $n \to \infty$,

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\triangle x \geq \lim_{n\to\infty} \sum_{i=1}^{n} g(x_i^*)\triangle x$$

i.e.

$$\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx.$$

□

Proof of Property (P7). Since $m \leq f(x) \leq M$, Property (P6) gives

$$\int_a^b m\,dx \leq \int_a^b f(x)\,dx \leq \int_a^b M\,dx.$$

Using Property (P1) to evaluate the integrals on the left and right sides, we have

$$m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a).$$

□

Though it's quite easy to write down the limit of Riemann sums given a definite integral, the reverse, i.e. converting a limit to a definite integral, may not be so straightforward.

Example 1.5. Express the following limit as a definite integral.

Final expressions are not unique.

(i) $\displaystyle \lim_{n\to\infty} \sum_{i=1}^{n} \frac{\pi^2 i}{n^2} \sin\left(\frac{i\pi}{n}\right)$

(ii) $\displaystyle \lim_{n\to\infty} \frac{3}{n} \sum_{i=1}^{n} \left(2 + \frac{3i}{n}\right)^4$

up to your choice

Solution. We will make use of right-endpoints approximations. The idea is to identify a suitable choice of $\triangle x$, $x_i = a + i\triangle x$, $f(x)$ (these choices may not be unique) which will determine the rest:

$$a = x_0, \quad b = x_n, \quad \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\triangle x = \int_a^b f(x)\,dx.$$

Strategy:

Identify Δx, $x_i = a + i\Delta x$

$\Delta x \to \frac{b-a}{n}$, $b-a \to$ constant

$a = x_0$, $b = x_n$

$\to f(x) \to \int_a^b f(x)\,dx$

* Wooclap Qn

$\sum_{i=0}^{n-1} f(x_i)\Delta x$ is the Riemann sum using

↳ left-hand point approximation

Started from zero!

Is it always true that $L_n \le A \le R_n$

↳ no. it depends if function is increasing or decreasing.

(i) Consider the i-th term:

$$\frac{\pi^2 i}{n^2}\sin\left(\frac{i\pi}{n}\right) = \frac{i\pi}{n}\sin\left(\frac{i\pi}{n}\right)\cdot\frac{\pi}{n}.$$

Choose

$$\triangle x = \frac{\pi}{n}, \quad x_i = \frac{i\pi}{n}, \quad f(x) = x\sin x.$$

Alternatively

$\Delta x = \frac{1}{n}$

$x_i = a + \Delta x$

$= a + \frac{i}{n}$

$x_i = \frac{i}{n}$ $\quad x_n = \frac{n}{n}$

$a = 0,\ b = 1$

$\sum_{i=1}^{n} \frac{i}{n}\pi\sin\left(\frac{i}{n}\pi\right)\pi\cdot\frac{1}{n}$

$= \sum_{i=1}^{n} \overbrace{x_i\pi\sin(x_i\pi)\pi}^{f(x_i)}\Delta x$

$= \sum_{i=1}^{n} \underbrace{\pi^2 x_i\sin(\pi x_i)}_{f(x) = \pi^2 x\sin(\pi x)}\Delta x$

$n\to\infty$ $\int_0^1 \pi^2 x\sin(\pi x)\,dx$

Then

$$a = x_0 = 0, \quad b = x_n = \pi.$$

So

$$\begin{aligned}
\lim_{n\to\infty}\sum_{i=1}^{n}\frac{\pi^2 i}{n^2}\sin\left(\frac{i\pi}{n}\right) &= \lim_{n\to\infty}\sum_{i=1}^{n}\frac{i\pi}{n}\sin\left(\frac{i\pi}{n}\right)\cdot\frac{\pi}{n}\\
&= \lim_{n\to\infty}\sum_{i=1}^{n} x_i\sin x_i\triangle x\\
&= \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\triangle x\\
&= \int_0^{\pi} f(x)\,dx\\
&= \int_0^{\pi} x\sin x\,dx
\end{aligned}$$

(ii) Consider the i-th term:

$$\left(2+\frac{3i}{n}\right)^4\cdot\frac{3}{n}.$$

Choose

$$\triangle x = \frac{3}{n}, \quad x_i = 2+\frac{3i}{n}, \quad f(x) = x^4.$$

Then

$$a = x_0 = 2, \quad b = x_n = 5.$$

So

$$\begin{aligned}
\lim_{n\to\infty} \frac{3}{n} \sum_{i=1}^{n} \left(2+\frac{3i}{n}\right)^4 &= \lim_{n\to\infty} \sum_{i=1}^{n} \left(2+\frac{3i}{n}\right)^4 \cdot \frac{3}{n} \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} (x_i)^4 \triangle x \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i) \triangle x \\
&= \int_2^5 f(x)\,dx \\
&= \int_2^5 x^4\,dx.
\end{aligned}$$

Can you find a different choice of $\triangle x$, x_i and $f(x)$? □

Example 1.6. Suppose f is continuous, and

$$\int_0^3 f(x)\,dx = 3, \qquad \int_0^4 f(x)\,dx = 7.$$

Find $\int_4^3 f(x)\,dx$.

Solution. By Property (P4),

$$\int_0^4 f(x)\,dx = \int_0^3 f(x)\,dx + \int_3^4 f(x), dx,$$

$$7 = 3 + \int_3^4 f(x)\,dx,$$

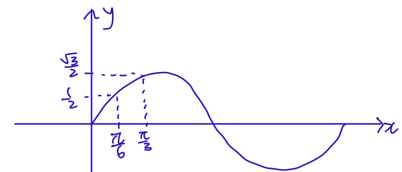
$$\int_3^4 f(x)\,dx = 4.$$

Hence,

$$\int_4^3 f(x)\,dx = -\int_3^4 f(x)\,dx = -4.$$

□

Example 1.7. Use the properties of integrals to verify that



$$\frac{\pi}{12} \leq \int_{\pi/6}^{\pi/3} \sin x \, dx \leq \frac{\sqrt{3}\pi}{12}.$$

$\therefore \frac{1}{2} \leq \sin x \leq \frac{\sqrt{3}}{2}$

Solution. Note that $\sin x$ is increasing on $[\pi/6, \pi/3]$, so for $\frac{\pi}{6} \leq x \leq \frac{\pi}{3}$ we have

$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{2} dx \leq \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin x \, dx \leq \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{3}}{2} dx$

$\underbrace{\frac{1}{2}\left(\frac{\pi}{3} - \frac{\pi}{6}\right)}_{\frac{\pi}{12}} \leq \quad '' \quad \leq \underbrace{\frac{\sqrt{3}}{2}\left(\frac{\pi}{3} - \frac{\pi}{6}\right)}_{\frac{\sqrt{3}}{12}\pi}$

$$\frac{1}{2} = \sin(\pi/6) \leq \sin x \leq \sin(\pi/3) = \frac{\sqrt{3}}{2}.$$

By Property P7, we have

$$\frac{1}{2}(\pi/3 - \pi/6) \leq \int_{\pi/6}^{\pi/3} \sin x \, dx \leq \frac{\sqrt{3}}{2}(\pi/3 - \pi/6).$$

$$\frac{\pi}{12} \leq \int_{\pi/6}^{\pi/3} \sin x \, dx \leq \frac{\sqrt{3}\pi}{2}.$$

□

Example 1.8. Which of the following integrals has the largest value? Why?

$$\int_1^2 \sqrt{x} \, dx, \quad \int_1^2 \sqrt{1/x} \, dx, \quad \int_1^2 \sqrt{\sqrt{x}} \, dx.$$

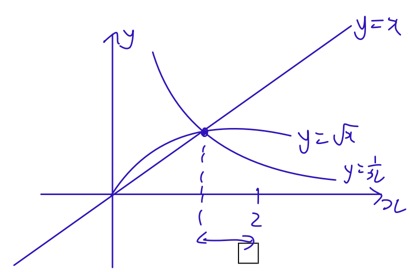
won't change the inequality

Solution. Note that $\frac{1}{x} \leq \sqrt{x} \leq x$ for $1 \leq x \leq 2$. Since $\sqrt{\cdot}$ is an increasing function, we have

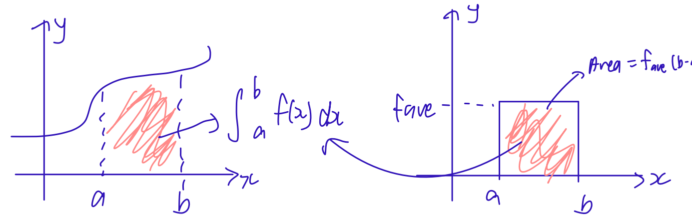
$$\sqrt{1/x} < \sqrt{\sqrt{x}} < \sqrt{x}, \quad 1 \leq x \leq 2.$$

So

$$\int_1^2 \sqrt{1/x} \, dx < \int_1^2 \sqrt{\sqrt{x}} \, dx < \int_1^2 \sqrt{x} \, dx.$$



Thus, $\int_1^2 \sqrt{x} \, dx$ has the largest value. □



Definition 3. (Average value)

We define the **average value** of f on $[a, b]$ as

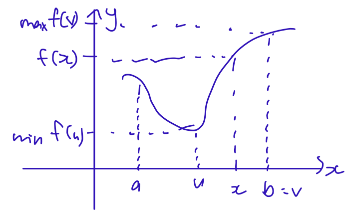
$$f_{ave} = \frac{1}{b-a} \int_a^b f(x)\,dx.$$

Theorem 2. (The Mean Value Theorem for Integrals)

If f is continuous on $[a, b]$, then there exists a number c in $[a, b]$ such that

$$f(c) = f_{ave} = \frac{1}{b-a} \int_a^b f(x)\,dx.$$

<u>Proof.</u> Since f is continuous on $[a, b]$, by the **Extreme Value Theorem**, there exist $u, v \in [a, b]$ such that



$$f(u) \le f(x) \le f(v), \quad \text{for all } x \in [a, b].$$

By Property (P1) and (P6), we have

$$(b-a)f(u) = \int_a^b f(u)\,dx \le \int_a^b f(x)\,dx \le \int_a^b f(v)\,dx = (b-a)f(v).$$

Dividing the terms by $(b-a)$, we have

f_{ave}

$$f(u) \le \frac{1}{b-a} \int_a^b f(x)\,dx \le f(v).$$

Finally, the **Intermediate Value Theorem** says that there must exist $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x)\,dx.$$

□

1.4 The Fundamental Theorem of Calculus

Reference: Stewart Calculus 8E, Section 4.3

Goal: To connect integration and differentiation, by showing that we can use antiderivative to calculate a definite integral (so that we can avoid using the limit of Riemann sums to evaluate a definite integral!)

Theorem 3. (Fundamental Theorem of Calculus Part 1 (FTC1))

Suppose f is continuous on $[a, b]$. Define

$$F(x) = \int_a^x f(t)\,dt, \quad a \leq x \leq b.$$

allow x to vary

Then $F(x)$ is differentiable on (a, b) with

$$F'(x) = \frac{d}{dx}\int_a^x f(t)\,dt = f(x), \quad a < x < b.$$

doesn't matter

In short: FTC1 says that "integration" followed by "differentiation" returns the original function!

Proof.

Idea: Use definition of differentiability to expand $F'(x)$. Then apply the Mean Value Theorem for Integrals (Theorem 2) to evaluate the resulting term to show that $F'(x) = f(x)$.

Step 1: Expand $F(x)$.

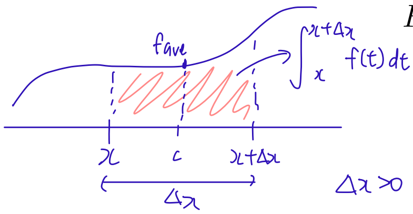
Let $x \in (a, b)$. Using the definition of differentiability, we have

$$F'(x) = \lim_{\triangle x \to 0} \frac{F(x + \triangle x) - F(x)}{\triangle x}. \tag{1.2}$$

0^+ or 0^-
it's two sided!

By the definition of $F(x)$, we have

$\frac{1}{\Delta x}\int_x^{x+\Delta x} f(t)\,dt = f_{ave}$ on $[x, x+\Delta x]$
$= f_c$



$\Delta x > 0$

$$\begin{aligned} F'(x) &= \lim_{\triangle x \to 0} \frac{\displaystyle\int_a^{x+\triangle x} f(t)\,dt - \int_a^x f(t)\,dt}{\triangle x} \\ &= \lim_{\triangle x \to 0} \frac{1}{\triangle x} \cdot \int_x^{x+\triangle x} f(t)\,dt \end{aligned} \tag{1.3}$$

transitive rule

<u>Step 2</u>: Use the Mean Value Theorem for Integrals.

Applying the Mean Value Theorem for Integrals (Theorem 2) to the function $f(t)$ on the interval $[x, x + \triangle x]$, there exists a number c, $x \le c \le x + \triangle x$ such that

$$f(c) = \frac{1}{\triangle x} \cdot \int_x^{x+\triangle x} f(t)\,dt. \tag{1.4}$$

Substituting (1.4) into (1.3):

c depends on Δx

$$F'(x) = \lim_{\triangle x \to 0} f(c). \tag{1.5}$$

Now, notice that $c \to x$ as $\triangle x \to 0$ (this is because c is sandwiched between x and $x + \triangle x$).

It follows from (1.5) that

$$F'(x) = \lim_{c \to x} f(c) = f(x),$$

as desired.

$\square$

Example 1.9. Find $\dfrac{dy}{dx}$ for $y = \displaystyle\int_1^x \sqrt{1+t^2}\,dt$.

Solution. By FTC1, we have

$$\frac{dy}{dx} = \frac{d}{dx}\left(\int_1^x \sqrt{1+t^2}\,dt\right) = \sqrt{1+x^2}.$$

1 → constant

→ original function in terms of x! not t!

□

Example 1.10. Find $\dfrac{d}{dx}\displaystyle\int_1^{x^4} \sec t\,dt$

no match ∴ need chain rule

Solution. We use Chain Rule in conjunction with FTC1. Let $u = x^4$. Then

$\frac{du}{dx} = 4x^3$

$$\begin{aligned}
\frac{d}{dx}\int_1^{x^4} \sec t\,dt &= \frac{d}{dx}\int_1^{u} \sec t\,dt \\
&= \frac{d}{du}\left(\int_1^{u} \sec t\,dt\right)\cdot\frac{du}{dx} \quad \text{(by Chain Rule)} \\
&= \sec u \cdot \frac{du}{dx} \quad \text{(by FTC1)} \\
&= \sec(x^4)\cdot 4x^3.
\end{aligned}$$

□

We now turn to the Fundamental Theorem of Calculus Part 2, which is the marriage between two seemingly unrelated concepts: definite integrals motivated by finding area under a curve, and antiderivatives motivated by reversing differentiation! In applications and higher dimensions, FTC2 also captures ideas like conservation of energy.

Theorem 4. (Fundamental Theorem of Calculus Part 2 (FTC2))

If f is continuous on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a) = [F(x)]_a^b$$

where F is any antiderivative of f.

Proof.

what we are claiming

Idea: Let $G(x) = \displaystyle\int_a^x f(t)\,dt$. Both $G(x)$ and $F(x)$ are antiderivatives of $f(x)$, and so they differ by a constant. We will exploit this fact.

from first lesson!

From FTC1, $G'(x) = f(x)$. So G is an antiderivative of f.

Given than F is any antiderivative of f. By Theorem 1,

$$F(x) - G(x) = C$$

where C is some constant.

Substituting $x = a$ and $x = b$, we have

$$G(a) = \int_a^a f(t)\,dt = 0, \quad G(b) = \int_a^b f(t)\,dt.$$

Thus,

$$\begin{gathered}(F(b) - G(b)) - (F(a) - G(a)) = C - C = 0 \\ \left(F(b) - \int_a^b f(t)\,dt\right) - (F(a) - 0) = 0 \\ \int_a^b f(t)\,dt = F(b) - F(a). \qquad \text{Magic or what!}\end{gathered}$$

□

Example 1.11. Evaluate the integral $\displaystyle\int_{-2}^1 x^3\,dx$

Solution.

$$\begin{aligned}\int_{-2}^1 x^3\,dx &= \left[\frac{x^4}{4}\right]_{-2}^1 \\ &= \frac{1}{4} - \frac{(-2)^4}{4} \\ &= -\frac{15}{4}.\end{aligned}$$

□

Example 1.12. Find the area of the region bounded between the cosine curve and the x-axis, from $x = 0$ to $x = \pi/2$.

Solution. The area is

$$\int_0^{\pi/2} \cos x \, dx = [\sin x]_0^{\pi/2} = 1.$$

$\square$

Example 1.13. Use the properties of integrals to verify that

$$\int_1^3 \sqrt{x^4+1}\,dx \geq \frac{26}{3}.$$

<u>*Solution.*</u> For $1 \leq x \leq 3$, we have $x^4+1 \geq x^4$, so $\sqrt{x^4+1} \geq \sqrt{x^4} = x^2$.

Thus,

$$\int_1^3 \sqrt{x^4+1}\,dx \geq \int_1^3 \sqrt{x^4}\,dx = \int_1^3 x^2\,dx = \left[\frac{x^3}{3}\right]_1^3 = \frac{26}{3}.$$

□

Example 1.14. Calculate the derivative

$$\frac{d}{dx}\int_{\sqrt{x}}^{x} \frac{\sin u}{u}\,du, \quad = \text{function of } x$$

To apply FTC 1, the lower limit needs to be a constant.

where $x \in (0,\infty)$. so $\frac{\sin u}{u}$ is continuous

<u>*Solution.*</u> Note that

$$\int_{\sqrt{x}}^{x} \frac{\sin u}{u}\,du = \int_{\sqrt{x}}^{a} \frac{\sin u}{u}\,du + \int_a^x \frac{\sin u}{u}\,du$$

for any $a \in (0,\infty)$.

we can also just leave a as it is.

Let's choose $a = 2$. Then the function $\frac{\sin x}{x}$ is continuous on $[\sqrt{x}, 2]$ and on $[2, x]$. By FTC1 and Chain Rule, we have

$$\begin{aligned}
\frac{d}{dx}\left(\int_{\sqrt{x}}^{x} \frac{\sin u}{u}\,du\right) &= \frac{d}{dx}\left(\int_{\sqrt{x}}^{2} \frac{\sin u}{u}\,du\right) + \frac{d}{dx}\left(\int_2^x \frac{\sin u}{u}\,du\right) \\
&= -\frac{d}{dx}\left(\int_2^{\sqrt{x}} \frac{\sin u}{u}\,du\right) + \frac{d}{dx}\left(\int_2^x \frac{\sin u}{u}\,du\right) \\
&= -\frac{d}{dw}\left(\int_2^{w} \frac{\sin u}{u}\,du\right)\frac{dw}{dx} + \frac{d}{dx}\left(\int_2^x \frac{\sin u}{u}\,du\right) \qquad (\text{where } w = \sqrt{x}) \\
&= -\frac{\sin w}{w}\frac{1}{2\sqrt{x}} + \frac{\sin x}{x} \\
&= -\frac{\sin\sqrt{x}}{\sqrt{x}}\frac{1}{2\sqrt{x}} + \frac{\sin x}{x} \\
&= \frac{\sin x}{x} - \frac{\sin\sqrt{x}}{2x}.
\end{aligned}$$

chain rule

differentiate $w=\sqrt{x}$

$\frac{dw}{dx}$

FTC

□

1.5 The Substitution Rule

replace x by $u = g(x)$

Reference: Stewart Calculus 8E, Section 4.5

> **Goal:** By changing the underlying variable x to a new variable u, we can convert a difficult integral with respect to x into one with respect to u that is easier to compute. This change of variable takes the form
>
> $$u = g(x),$$
>
> where g is some function of x.

"Chain Rule" for integration

> **Theorem 5. (Substitution Rule for Definite Integral)**
>
> Suppose g' is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$. Then
>
> $$\int_a^b f(g(x))\underbrace{g'(x)}_{\frac{du}{dx}}\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

Proof. Let F be an antiderivative of f. Notice that $\frac{d}{dx}(F(g(x)) = F'(g(x))g'(x) = f(g(x))g'(x)$ by Chain Rule. This means that $F(g(x))$ is an antiderivative of $f(g(x))g'(x)$. Thus, by FTC2,

need to show

equals

$$\int_a^b f(g(x))g'(x)\,dx = F(g(b)) - F(g(a)). \tag{1.6}$$

On there other hand, since F is an antiderivative of f, we have (by FTC2)

$\int_a^b f(x)\,dx = F(b) - F(a)$

$$\int_{g(a)}^{g(b)} f(u)\,du = F(g(b)) - F(g(a)). \tag{1.7}$$

Combining (1.6) and (1.7), we have

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du,$$

as desired. □

$$\begin{aligned}\frac{d}{dx}F(\overbrace{g(x)}^{u}) &= \frac{d}{du}F(u)\cdot\frac{du}{dx} \quad [\text{Chain Rule}]\\ &= F'(u)\cdot g'(x)\\ &= f(u)\cdot g'(x)\\ &= f(g(x))\cdot g'(x)\end{aligned}$$

In practice, when applying the Substitution Rule to evaluate $\int_a^b h(x)\,dx$, we do the following:

- Think of a function $u = g(x)$. Then $\dfrac{du}{dx} = g'(x)$. From here, try to write $dx = k(u)\,du$ for some function k of u.

  $\to du = g'(x)dx$

- Then try to convert the integral $\int_a^b h(x)\,dx$ with respect to x to an integral with respect to u.
- Replace the lower limit a by $g(a)$, and the upper limit b by $g(b)$ when computing the resulting integral with respect to u.

Example 1.15. Evaluate $\displaystyle\int_1^2 \frac{1}{(3-5x)^2}\,dx$.

Solution.

- Let $u = g(x) = 3 - 5x$. Then $\dfrac{du}{dx} = g'(x) = -5 \Longrightarrow dx = \dfrac{1}{(-5)}\,du$
- Then
$$\int \frac{1}{(3-5x)^2}\,dx = \int \frac{1}{u^2}\cdot\frac{1}{(-5)}\,du = \frac{1}{(-5)}\int \frac{1}{u^2}\,du$$

Substituting the lower and upper limits:

$$\begin{aligned}
\int_1^2 \frac{1}{(3-5x)^2}\,dx &= \frac{1}{(-5)}\int_{g(1)}^{g(2)} \frac{1}{u^2}\,du \\
&= \frac{1}{(-5)}\int_{-2}^{-7} \frac{1}{u^2}\,du \\
&= -\frac{1}{5}\left[-u^{-1}\right]_{-2}^{-7} \\
&= -\frac{1}{5}\left(\frac{1}{7}-\frac{1}{2}\right) \\
&= \frac{1}{14}.
\end{aligned}$$

□

Example 1.16. Evaluate $\int_0^4 \sqrt{2x+1}\,dx$.

Solution.

- Let $u = g(x) = \sqrt{2x+1}$. Then $du = \frac{1}{2}(2x+1)^{-1/2}(2)\,dx \implies dx = u\,du$.
- Then
$$\int \sqrt{2x+1}\,dx = \int u(u\,du) = \int u^2\,du.$$

Substituting the lower and upper limits:

$$\begin{aligned}\int_0^4 \sqrt{2x+1}\,dx &= \int_{g(0)=1}^{g(4)=3} u^2\,du \\ &= \left[\frac{u^3}{3}\right]_1^3 \\ &= 9 - \frac{1}{3} \\ &= \frac{26}{3}.\end{aligned}$$

□

Bonus Example:

$$\int_0^1 e^{\sqrt{x}}\,dx$$ makes it difficult to integrate

☆ key learning point in this example

still need to convert to u!

let $u = \sqrt{x}$ $\quad \frac{du}{dx} = \frac{1}{2\sqrt{x}}$ $\quad dx = 2\sqrt{x}\,du = 2u\,du$

$$\therefore \int_{0 \to \sqrt{0}}^{1 \to \sqrt{1}} e^u \cdot 2u\,du$$ → can do integration by parts in the future

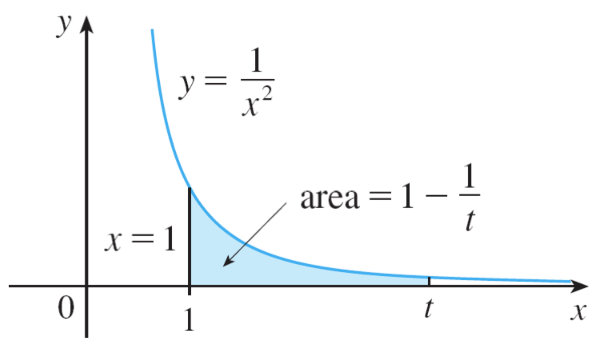
1.6 Improper Integrals

Reference: Stewart Calculus 8E, Section 7.8

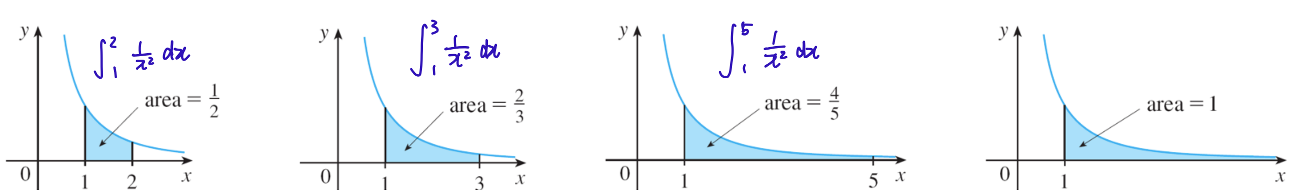
Goal: Previously, in defining a definite integral $\int_a^b f(x)\,dx$ we dealt with function f which is continuous on closed and bounded interval $[a,b]$. In this section, we wish to extend the notion of integrals to cases where

- the underlying interval is infinite, e.g. $[a,\infty)$, $(-\infty,b]$, $(-\infty,\infty)$
- the integrand f has an infinite discontinuity on a finite interval.

Motivation: Suppose we want to compute $\displaystyle\int_1^\infty \frac{1}{x^2}\,dx$, which is the area under the curve $y = \dfrac{1}{x^2}$, above the x-axis, and on the right-hand side of the line $x = 1$:



Idea: We can first calculate the area between $x = 1$ and $x = t$ (for some t). As t gets larger and larger, the area seems to converge to the area we want.



This brings us to the following definition of improper integrals.

$$A = \lim_{t\to\infty}\int_1^t \frac{1}{x^2}\,dx$$

underlying intervals is infinite

Definition 4. (Improper Integral of Type 1)

(a) If $\int_a^t f(x)\,dx$ exists for every number $t \geq a$, then

$$\int_a^\infty f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx.$$

(b) If $\int_t^b f(x)\,dx$ exists for every number $t \leq b$, then

$$\int_{-\infty}^b f(x)\,dx = \lim_{t\to-\infty} \int_t^b f(x)\,dx.$$

In each case, the improper integrals are called **convergent** if the corresponding limit exists, and **divergent** if the limit does not exist.

if only 1 diverges, the entire $\int_{-\infty}^{\infty} f(x)\,dx$ diverges

(c) If both $\int_a^\infty f(x)\,fx$ and $\int_{-\infty}^a f(x)\,dx$ are convergent, then we define

arbitrary

$$\int_{-\infty}^\infty f(x)\,dx = \underbrace{\int_{-\infty}^a f(x)\,dx}_{\text{case b}} + \underbrace{\int_a^\infty f(x)\,dx}_{\text{case a}}.$$

Note: In part (c), any real number a can be used. This will be justified in the tutorial (Tutorial 3).

infinite interval

Example 1.17. Determine whether $\int_1^\infty \frac{1}{x}\,dx$ is convergent or divergent.

Solution. Let $t \geq 1$. Then

x is always +ve so technically can drop the absolute sign

$$\int_1^t \frac{1}{x}\,dx = [\ln|x|]_1^t = \ln t,$$

exists. Now,

$$\int_1^\infty \frac{1}{x}\,dx = \lim_{t\to\infty} \int_1^t \frac{1}{x}\,dx = \lim_{t\to\infty} \ln t = \infty.$$

Thus, the improper limit is divergent. ⇒ dne

□

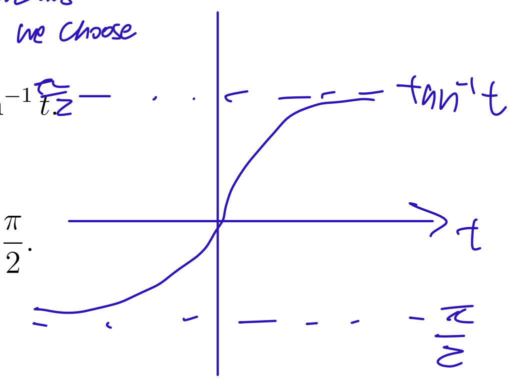
need to split the interval!

Example 1.18. Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$.

Solution. Let $a = 0$ in part (c) of Definition (4). Then

will get same ans no matter what value of a we choose

$$\int_t^0 \frac{1}{1+x^2}\,dx = \left[\tan^{-1} x\right]_t^0 = \tan^{-1} 0 - \tan^{-1} t = -\tan^{-1} t.$$



So

$$\lim_{t\to-\infty} \int_t^0 \frac{1}{1+x^2}\,dx = \lim_{t\to-\infty} \left(-\tan^{-1} t\right) = -\left(-\frac{\pi}{2}\right) = \frac{\pi}{2}.$$

On the other hand,

$$\int_0^t \frac{1}{1+x^2}\,dx = \left[\tan^{-1} x\right]_0^t = \tan^{-1} t - \tan^{-1} 0 = \tan^{-1} t.$$

So

$$\lim_{t\to\infty} \int_0^t \frac{1}{1+x^2}\,dx = \lim_{t\to\infty} \tan^{-1} t = \frac{\pi}{2}.$$

Combining, we have

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \lim_{t\to-\infty} \int_t^0 \frac{1}{1+x^2}\,dx + \lim_{t\to\infty} \int_0^t \frac{1}{1+x^2}\,dx = \frac{\pi}{2} + \frac{\pi}{2} = \pi.$$

$\square$

WARNING: The following calculations are WRONG!

not actual numbers!

$\infty - \infty$

(i) $\displaystyle\int_{-\infty}^{\infty} x\,dx = \left[\frac{x^2}{2}\right]_{-\infty}^{\infty} = 0.$ even if somehow get correct ans, it is illegal

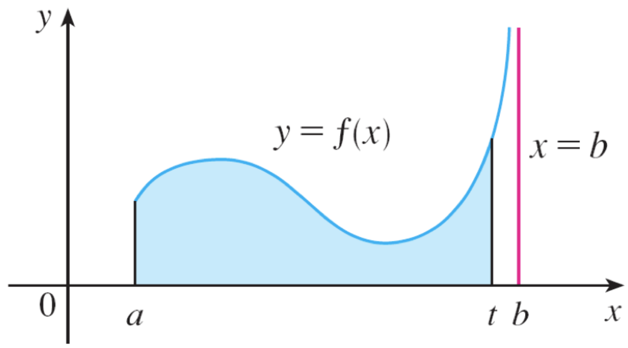
(ii) $\displaystyle\int_{-\infty}^{\infty} x\,dx = \lim_{t\to\infty} \int_{-t}^{t} x\,dx = \lim_{t\to\infty} \left[\frac{x^2}{2}\right]_{-t}^{t} = \lim_{t\to\infty} \left(\frac{t^2}{2} - \frac{(-t)^2}{2}\right) = \lim_{t\to\infty} 0 = 0.$

not the correct definition, supposed to split up the integral

What is the correct way to determine $\displaystyle\int_{-\infty}^{\infty} x\,dx$?

should be diverging, dne.

An integral can also be 'improper' on a finite interval. For example, this happens when the function has an asymptote at the endpoint of the interval, or even at a point in the middle of the interval.



Definition 5. (Improper Integral of Type 2)

(a) If f is continues on $[a, b)$ and is discontinuous at b, then

$$\int_a^b f(x)\,dx = \lim_{t \to b^-} \int_a^t f(x)\,dx$$

if this limit exists (as a finite number).

(b) If f is continuous on $(a, b]$ and is discontinuous at a, then

$$\int_a^b f(x)\,dx = \lim_{t \to a^+} \int_t^b f(x)\,dx$$

if this limit exists (as a finite number).

In each case, the improper integrals are called **convergent** if the corresponding limit exists, and **divergent** if the limit does not exist.

(c) If f has a discontinuity at c, where $a < c < b$, and both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ are convergent, then we define

$$\int_a^b f(x)\,dx = \underbrace{\int_a^c f(x)\,dx}_{\text{case a}} + \underbrace{\int_c^b f(x)\,dx}_{\text{case b}}.$$

Example 1.19. Find $\displaystyle\int_2^5 \frac{1}{\sqrt{x-2}}\,dx$.

Solution. The function $f(x) = \dfrac{1}{\sqrt{x-2}}$ has an infinite discontinuity at the left endpoint of $(2,5]$. By Definition 5,

$$\int_2^5 \frac{1}{\sqrt{x-2}}\,dx = \lim_{x\to 2^+} \int_t^5 \frac{1}{\sqrt{x-2}}\,dx$$

Substituting $u = x-2$, we have

$$\int \frac{1}{\sqrt{x-2}}\,dx = \int \frac{1}{\sqrt{u}}\,du = 2\sqrt{u} + C = 2\sqrt{x-2} + C.$$

Thus,

$$\begin{aligned}\int_2^5 \frac{1}{\sqrt{x-2}}\,dx &= \lim_{t\to 2^+} \left[2\sqrt{x-2}\right]_t^5 \\ &= \lim_{t\to 2^+} 2(\sqrt{3} - \sqrt{t-2}) \\ &= 2\sqrt{3}.\end{aligned}$$

converges, limit exists

□

Example 1.20. Evaluate $\displaystyle\int_0^3 \frac{1}{x-1}\,dx$ if possible.

Solution. The function $f(x) = \dfrac{1}{x-1}$ has a discontinuity at $x = 1$ inside the interval $[0,3]$. By part (c) of Definition 5, the integral (if it exists) is equal to

$$\int_0^3 \frac{1}{x-1}\,dx = \int_0^1 \frac{1}{x-1}\,dx + \int_1^3 \frac{1}{x-1}\,dx,$$

where both $\displaystyle\int_0^1 \frac{1}{x-1}\,dx$ and $\displaystyle\int_1^3 \frac{1}{x-1}\,dx$ must exist.

check both of this

Unfortunately, $\int_0^1 \frac{1}{x-1}\,dx$ does not exist because

$$\begin{aligned}
\int_0^1 \frac{1}{x-1}\,dx &= \lim_{t\to 1^-}\int_0^t \frac{1}{x-1}\,dx \\
&= \lim_{t\to 1^-}\left[\ln|x-1|\right]_0^t \\
&= \lim_{t\to 1^-}\left(\ln|t-1| - \ln|0-1|\right) \\
&= \lim_{t\to 1^-}\ln|t-1| \\
&= \lim_{t\to 1^-}\ln(1-t) \\
&= -\infty.
\end{aligned}$$

no need to check the other one cos both need to be convergent for it to exist.

Hence, the integral $\int_0^3 \frac{1}{x-1}\,dx$ is divergent. $\square$

WARNING: The following calculation is WRONG!

$$\int_0^3 \frac{1}{x-1}\,dx = \left[\ln|x-1|\right]_0^3 = \ln 2 - \ln 1 = \ln 2.$$

This is wrong because the integral is improper and must be calculated in terms of limits as given in Definition 5.

* must always check if it is continuous at the interval

so we know if it is improper